

Touch & Discover

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Figure 1: Touch & Discover with focus on Ingemar Johansson

ABSTRACT

Touch & Discover is an interactive museum console developed by five interaction design students to enrich the visitor experience at the Gothenburg Sports Museum, *Idrottsmuseet*. The prototype focuses on the life and achievements of legendary Swedish boxer Ingemar Johansson and aims to address the museum's challenges of cluttered displays and limited information. By combining a touch-screen interface with curated multimedia content and real-world exhibit interaction through light cues, the project enhances accessibility, storytelling, and engagement. Although created as a prototype, the system demonstrates how interactive technology can transform static exhibitions into immersive and educational experiences for a broader audience.

KEYWORDS

interaction, museum, console, touch-screen

1 INTRODUCTION

The Gothenburg Sports Museum, *Idrottsmuseet* [4], consists of large collections of over 15 000 items and around 30 000 photographs [4]. The exhibits are installed randomly, making it difficult to reach some of them. Therefore, the elements need to be organized in an official manner. However, currently their visitors consist mainly of middle-aged men, and the museum wants to become more attractive to a larger audience by making it more interactive.

One such attempt will be described in this report, where five interaction design students set out on a journey to create a more interactive experience at the museum. This project revolved around Ingemar Johansson, but it could be done for any athlete, or other subject.

The project consists of a touch-screen based interface that allows users to explore various categories and find more information about an athlete. It aims to demonstrate how interactivity and thoughtful design can improve accessibility, storytelling, and curiosity in museums.

1.1 Background

Idrottsmuseet is situated in Kviberg, Gothenburg. It started in 1983 and contains physical editions of Swedish sports magazines from back in the 1980s [7]. At the moment the museum is split over three floors with artefacts on every surface. With all the artefacts there is almost no space for information plaques. This makes it hard to learn about the artefacts that a visitor doesn't already know about if they don't have a knowledgeable person guiding them.

Two of the biggest problems that were noticed when visiting the museum are; the overly cluttered exhibits as well as the lack of information about the artefacts. There are hundreds of items grouped together without a clear purpose. For example; is it because they are the same size? or because they belong to the same person? Which makes the space feel unorganized and cluttered. But the more pressing problem is the lack of readily available information, which hinders visitors from learning about the exhibits, and the athletes in general. Which might be a reason why there are mostly

*All authors contributed equally to this project.

middle-aged men visiting the museum, because they already possess knowledge of the athletes and their artefacts. The exhibits don't appear to be made for learning, the museum appears more like a place to simply store artefacts. It lacks fundamental aspects to let visitors learn about the athletes on their own without a guide to follow them around, explaining the artefacts.

To create the project, information about Ingemar Johansson has been used to populate the prototype. Ingemar Johansson was one of Sweden's most iconic athletes, but his story remains largely untold in interactive formats. *Idrottsmuseet* currently displays several artefacts from his life and career; however, these are mostly accompanied by static text that fails to capture the attention of many visitors. The lack of interactivity and narrative cohesion results in a museum experience that can feel fragmented, particularly to younger audiences or those unfamiliar with the sport of boxing.

1.2 Purpose

The purpose of the *Touch & Discover* console is to help visitors learn more about the exhibits in an interactive way, rather than relying on traditional text panels. Watching videos and figuring out which exhibits relate to the content will enrich the visitor experience and draw more attention to the museum's elements.

Visitors need guidance to find information about each exhibit. To enhance the museum experience, the elements should be organized and linked to relevant information. Additionally, visual design should attract the eye and give visitors a reason to stop and engage with the exhibits.

Storytelling is also a fun and engaging way to teach visitors about the exhibits. Currently, the exhibits are arranged somewhat randomly, and their information is hidden or hard to access. They need a guider to help tell their stories.

Accessibility is another important aspect. Reaching and understanding the exhibits can be challenging for some visitors, so we need to design with inclusivity in mind — making sure the experience is accessible to everyone, not just a few.

We believe that installing touch-screens will help the exhibits tell their stories while also saving space in the museum, eliminating the need for printed descriptions next to each exhibit.

2 RELATED WORK

Several museums have successfully implemented interactive technologies to enhance visitor engagement and learning. For example, Universeum [8] and Världskulturmuseet [9] in Gothenburg incorporate interactive consoles as a core part of their exhibitions. These consoles allow visitors to explore content at their own pace, promoting active participation rather than passive observation. In particular, Världskulturmuseet makes extensive use of headphones placed near screens, enabling an immersive storytelling experience. These audio narratives provide personal, cultural, or historical perspectives that deepen the visitor's understanding of the exhibition themes.

Similarly, the Museum of London [6] employs touch-screen displays that allow users to navigate digital archives containing historical documents, photographs, and videos. These interactive stations are contextually tied to the physical exhibits, giving users an opportunity to explore additional layers of information and make

connections that might not be immediately apparent from the physical artefacts alone. This approach supports both casual browsing and more in-depth exploration, catering to different levels of visitor interest and engagement.

These examples highlight the growing trend in museums to integrate multimedia and interactive elements as a way to enrich the visitor experience and make cultural and historical content more accessible, engaging, and personalized.

3 DESIGN & IMPLEMENTATION

The design process started with ideation which included sketching and making decisions about the design as a whole, and was later split into working separately on the physical and the digital parts of the design.

3.1 Ideation

The ideation phase started with the group communicating what aspects they wanted to be included, which then continued into paper sketches.

Our group drew various different sketches, each sketch included a potential solution for the problem. We used other methods for sketching than just drawing. For example, we wrote some ideas in form of text. In the sketching phase, we gave each member five minutes to draw, write or think about the idea. After finishing, we collected all ideas and presented them to each other. Lastly we voted for the best idea, and our vote landed on a touch-screen idea, as seen in Figure 2.

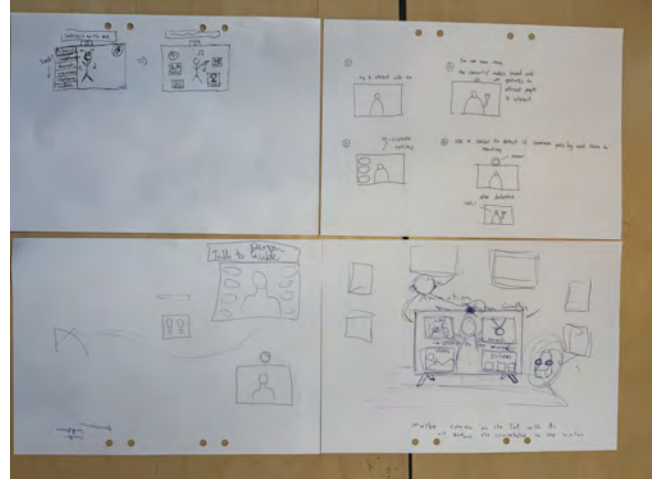


Figure 2: Initial sketches

To present our idea better, and discuss it with the supervisor, we created a digital sketch by using Adobe Photoshop [1]. The sketch is of a touch-screen. On the screen is Ingemar Johansson in the middle and around him are icons representing the information that will be presented, see figure 3.

These sketches were iterated until a basic design was agreed upon. This design consisted of a touch-screen and a physical component that would frame said screen, which would allow the user to navigate between information about the subject. At which point

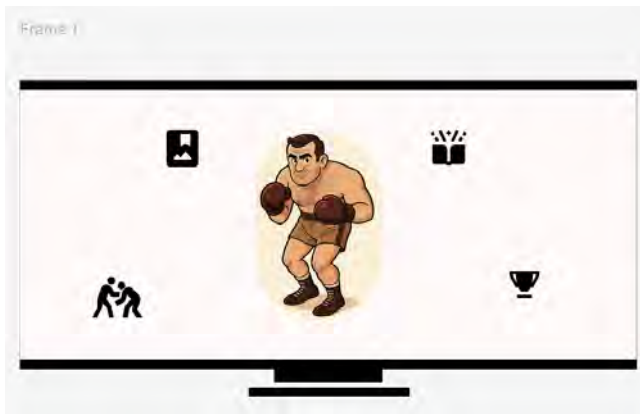


Figure 3: A digital sketch showing the initial design

the project's different parts were split among the group, the biggest segregation being between the digital and the physical.

3.2 Digital Design

The digital interface was mainly created using the program Intuiface [5].

Figma [3] was used to get a concept design made quickly. The design had a navigation bar at the bottom with five items on it, and a basic video above as the rest state, see Figure 4.

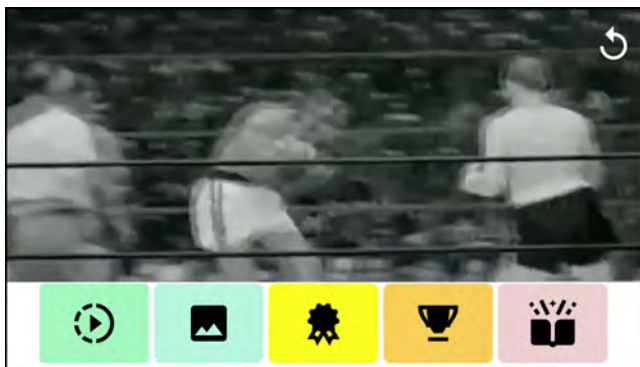


Figure 4: Figma prototype of the interface

When this was moved over to Intuiface, and implementation began, a few alterations were made over the course of the project. The navigation bar was slimmed down to three icons because of lack of time to implement all of them. While the remaining pages were designed and built.

The first page contains videos about the subject, arranged in a carousel with a start and end point. The videos used to populate this page are both clips from Ingemar Johansson's matches as well as short documentaries about aspects of his life.

The second page focuses on allowing the user to view photos related to the subject, arranged in a grid with a vertical scroll. When a photo is clicked it gets superimposed on the screen and buttons at the edges allow for easy navigation between images. Each image

also comes with a short information paragraph that describes it and who took the picture.

The last page is a world map showing important places related to the subject. This page focuses on awards and where he got them. Each place Ingemar Johansson has won something in is highlighted with a red dot on the map. When interacting with the dot, the background will darken to draw attention to the pop-up. Said pop-up will contain information about the award connected to the place.

Figure 5 shows the latest iteration of the design. The image on screen is the rest screen for the interface.

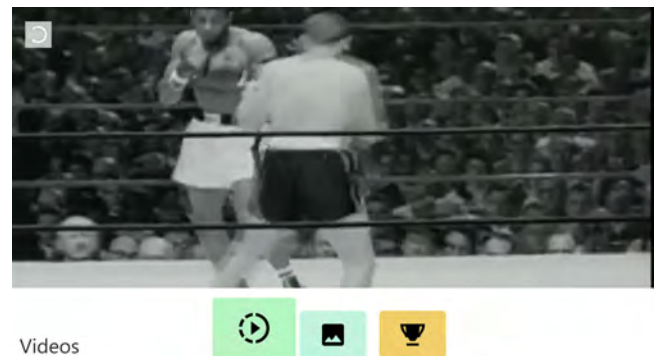


Figure 5: The rest screen for the final digital design

The latest iteration of the prototype connects the information on the screen to the physical exhibit around it. This is done through a light turning on over the real object, helping the user to find it more easily. Said light turns on when the corresponding item is highlighted on the screen.

3.3 Physical Design

The physical design includes everything about the scenery around the digital design.

The original idea was to make a simple frame but as time went on it evolved to instead focus on a way to connect the digital design to the physical environment. This was done by lighting up the exhibits whenever the user interacted with them on the screen. The screen would be positioned in the middle and the various objects on shelves around it. Furthermore, headphones would be available for the users so they could listen to the content without disrupting other visitors in the room.

Prototyping for this scenery was created using a large board. A background with shelves and artefacts related to Ingemar Johansson were plotted and stapled to the board. To prototype the lights, six LED-strips and six buttons were programmed using two Arduino Unos [2]. This allowed for the use of Wizard-of-Oz technique to cause the LED-strips to light up when users looked at certain things on the screen. Six lightbulbs were also 3D-printed using transparent filament and hot-glued to the board, covering the LED-strips but letting some light through. The result of this can be seen in Figure 7.

4 RESULT & DISCUSSION

The *Touch & Discover* prototype was set up in *Idrottshögskolan* one hour before the presentation started. The screen was mounted on a stand and positioned on a round table, see Figure 6.



Figure 6: The digital touch-screen

Behind the screen, we installed a board with a printed image of a museum shelf for Ingemar Johansson's section. On each shelf in the image, we placed photos of physical exhibits related to Ingemar Johansson, such as his championship belt or a portrait of him. Above each photo, we installed a small light. When a user selects a digital element on the screen - such as a video of Johansson's fight with Patterson - the corresponding light above the related physical exhibit turns on. For example, if the visitor watches the video of that specific match, the light above the photo of that match will illuminate. This creates a link between the digital content (e.g., the video) and the physical elements (e.g., the exhibit photo), enhancing the visitor's experience. Each digital element also includes a reference to the shelf number where the related physical item is located, see Figure 7.

We faced some challenges while presenting the prototype. We installed buttons for each shelf to manually turn the lights on and off based on the selected digital content. Since we couldn't directly program the screen to control the lights, we created a separate prototype using Arduino. We placed six buttons on the sides of the board, three on the right and three on the left. However, the buttons were mounted quite high, making them difficult to reach while visitors were engaging with the screen.

Despite this, visitors responded positively to the prototype. Many said we successfully connected digital technology with physical elements in an interactive way, which matched what our stakeholders hoped to achieve. We also made hidden information more visible. Some details in the museum were previously overlooked, making the experience less engaging. Our prototype brought these stories to light, making the content clearer and more appealing.

The prototype experience is designed to be eye-catching. As visitors enter the Ingemar Johansson section, the video presentation



Figure 7: The screen in front of the shelf

immediately draws their attention, encouraging them to stop, learn more about him, and begin exploring the related exhibits.

We are satisfied with our prototype and how far we have gotten. We are aware that we could have spent more of our time on explaining our idea within the group. Many times we found small inconsistencies between our own images of what we were supposed to do. Something that wasn't helped in how we split our work between the group members. The overall concept is something we believe that *Idrottsmuseet* could have great use of. We hope that if the museum chooses to implement this then the exhibits will be more accessible and interesting for those that want to learn.

A limitation of our prototype is that the touchscreen can get dirty or become less responsive if many people use it, which might make it harder to interact with. Another issue is that the design may not work well for everyone. For example, people with colour blindness or poor eyesight might struggle to see certain parts of the screen, especially if too much depends on colour or small text. Right now, there is no support for voice navigation, which could also make it less accessible for some users. The setup also requires visitors to stand close to the screen, which might not be comfortable for everyone, especially those with limited mobility. Lastly, even though we plan to use headphones for the sound, this could bring other challenges, such as hygiene concerns, wear and tear, or not fitting comfortably for all users.

5 FUTURE WORK

While the *Touch & Discover* prototype effectively demonstrates how interactivity can enhance the museum experience, there are several areas that could be explored further in future iterations:

- **Automatic Exhibit Highlighting:** The current prototype uses manually operated buttons to light up corresponding exhibits. The final product would use a fully integrated system where the touch-screen interface automatically triggers the lights to improve usability.

- **Expanded Content and Scalability:** Although the prototype focuses on Ingemar Johansson, the system could be adapted to cover other athletes, sports, or historical figures. A modular content management approach would make it easier for museums to update or repurpose the system for various exhibitions.
- **Formal User Testing and Evaluation:** Structured user testing, including usability studies, interviews, and observational research, is essential to evaluate how visitors interact with the system and what they learn. This feedback would provide data-driven insights for further design improvements.
- **Inclusive and Multilingual Design:** To reach a broader audience, future versions should support multiple languages, accessible design principles (e.g., screen readers, high contrast modes), and interaction options tailored for visitors with motor or cognitive impairments.

6 CONCLUSION

The *Touch & Discover* project successfully demonstrates how interactive technology can improve visitor engagement and storytelling in a museum context. By combining a digital touch-screen interface with physical exhibits, the prototype bridges the gap between static displays and immersive experiences. Visitors not only receive curated information through video, audio, and text, but are also guided to the corresponding real-life artefacts, making the visit process more attractive and enjoyable.

Although no formal user testing or visitor feedback was conducted, the prototype functions as a strong proof of concept. It addresses key issues such as information discoverability, exhibit accessibility, and the lack of interactivity often found in traditional displays. The project also shows potential for adaptation to other museum topics and exhibitions beyond Ingemar Johansson.

In conclusion, *Touch & Discover* illustrates how carefully designed interactive systems can transform static museum environments into engaging and educational experiences. Future iterations could focus on technical refinements and user testing to validate its impact more thoroughly.

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